

## All Reviews of cec\_pap158 [details](#)

### Decision: Accept

**Submission:** Instance-Aware Parameter Configuration in Bilevel Late Acceptance Hill Climbing for the Electric Capacitated Vehicle Routing Problem

**Contributors:** Bazargani, Chen, Qin, Wang

Key for the below column headings [show](#)

### Summary of Reviews of cec\_pap158s1

| Reviewer                   | rec <small>i</small> | conf <small>i</small>  | oral <small>i</small> | bp <small>i</small> | indust <small>i</small> |
|----------------------------|----------------------|------------------------|-----------------------|---------------------|-------------------------|
| <a href="#">Reviewer 1</a> | accept (4)           | somewhat confident (2) | oral (1)              | No (0)              | Yes (1)                 |
| <a href="#">Reviewer 2</a> | weak accept (3)      | confident (3)          | poster (0)            | No (0)              | No (0)                  |
| <a href="#">Reviewer 3</a> | accept (4)           | somewhat confident (2) | poster (0)            | No (0)              | No (0)                  |
| <b>Averages:</b>           | <b>3.7</b>           | <b>2.3</b>             | <b>0.3</b>            | <b>0.0</b>          | <b>0.3</b>              |

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## Full Reviews

### Reviewer 1

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#### ► Novelty and Significance

monospace

The proposed approach is novel in that it uses predicts the appropriate parameters to apply on instances of the E-CVRP. This approach presents a customised approach for solving each instance of the problem as opposed to cycling the whole dataset which can be time consuming. This is significant in reducing manhours of configuration and increasing the objective function.

#### ► Relevance

monospace

The paper is relevant. It presents relevant work and also in an appropriate problem domain.

#### ► Rigour and Soundness

monospace

The proposed approach is compared to a baseline study that was configured used Itrace which is a sound approach. However, to justify rigour it would've been good to use another dataset with different characteristics. It is important to evaluate the algorithm's ability to generalise. Scalability is another trait that should have been evaluated by instances with larger number of characteristics.

#### ► Structure and Readability

monospace

The paper is well structured and written with clarity making it readable.

## Invite to TEVC

monospace

no

If so, why and what extensions would you recommend?

## Review to TEVC

### ► Other Comments for Authors

markdown

monospace

1. Elaboration is required on the following statement ...However, E-CVRP instances exhibit pronounced heterogeneity in terms of network topology, customer demand distributions, and energy-related constraint.
2. How did the authors arrive at ridge regression as the choice of predictor? Was it an intuitive selection-based characteristics or other approaches were considered and tested?

### Scored Review Questions

|              | rec        | conf                   | oral     | bp     | indust  |
|--------------|------------|------------------------|----------|--------|---------|
| cec_pap158s1 | accept (4) | somewhat confident (2) | oral (1) | No (0) | Yes (1) |

## Reviewer 2

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### ► Novelty and Significance

monospace

The paper presents an instance-aware parameter configuration study for a heuristic (LAHC) for a specific VRP.

Although the approach itself is not particularly novel (its methodology for instance-aware parameter configuration is relatively standard), it applies this technique for an algorithm/problem pair where it was not done before.

The results (in terms of the improvement of the configuration scheme over the baseline), although statistically significant, are relatively minor.

### ► Relevance

monospace

The work fits well with the CECSS29 track and studies a topic that is highly relevant for the EC community.

### ► Rigour and Soundness

markdown

monospace

The overall structure of the study is reasonable. However, there are some details that I find need better justification/clarification:

- The LAHC heuristic has 5 parameters, but only 2 were chosen for the tuning study. The justification in the paper is that these two "... have been identified as the most influential in previous sensitivity analyses". How this analysis was performed, and its detailed results, are not provided.
- During the two-stage tuning strategy, over 10,000 configurations were tested. The searched parameter space has only 2 dimensions. This, to me, seems to be a rather excessive effort.
- Only 10 independent runs were performed per instance. With such a small sample, the confidence in the obtained results is not very high.
- The importance of the different features of the instances on the proposed parametrization is described only vaguely and only for the `L_h` parameter.
- Neither code nor data from the experiments were provided, which further limits the possibility of other researchers building upon the work and also limits the trust in the obtained results.

## ► Structure and Readability

monospace

The readability of the paper is without any major problems.

## Invite to TEVC

monospace

no

If so, why and what extensions would you recommend?

## Review to TEVC

## ► Other Comments for Authors

None

### Scored Review Questions

|              | rec ⓘ           | conf ⓘ        | oral ⓘ     | bp ⓘ   | indust ⓘ |
|--------------|-----------------|---------------|------------|--------|----------|
| cec_pap158s1 | weak accept (3) | confident (3) | poster (0) | No (0) | No (0)   |

## Reviewer 3

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## ► Novelty and Significance

markdown

monospace

### 1. Novelty: Moderate

The paper explores **Instance-Aware Parameter Configuration (IAPC)** applied to the Electric Capacitated Vehicle Routing Problem (E-CVRP).

- **Application-Specific Novelty:** While the concept of Per-Instance Algorithm Configuration (PIAC) is well-established in the broader AI community (e.g., SATzilla for SAT, ISAC for MIP), its specific application to the E-CVRP using the b-LAHC metaheuristic is relatively unexplored. The authors correctly identify a gap in the literature regarding the intersection of E-CVRP solvers and automated per-instance tuning.
- **Methodological Novelty:** The underlying techniques used—**IRACE** for offline label generation and **Ridge Regression** for prediction—are standard tools. There is no significant algorithmic innovation in the machine learning component itself. However, the **"gap-based multi-modality analysis"** (Section IV-B) used to process the training labels and handle non-convex parameter landscapes is a thoughtful, albeit minor, methodological contribution tailored to the noise inherent in metaheuristic tuning.
- **Feature Engineering:** The adaptation of extensive graph-theoretic features (kNN graphs, MST, Laplacian eigenvalues) specifically for E-CVRP instance characterization is a strong point, though largely derived from existing literature on fitness landscape analysis.

### 2. Significance: Moderate to High

The work holds practical significance for the operations research community, though the performance gains are incremental.

- **Robustness over Performance:** The reported average improvement of **0.28%** over the globally tuned baseline is statistically significant but relatively small in absolute terms. However, the significance lies in the **consistency** of the approach (improving 7 out of 8 test instances). This demonstrates that dynamic configuration is a viable path for improving robustness without changing the core solver code.
- **Interpretability:** A significant contribution is the **Feature Importance Analysis** (Section V-D.4). By correlating instance features (e.g., demand entropy, spectral properties) with optimal parameter settings ( $\$L\_h\$$ ), the paper provides scientific insight into *why* certain instances are harder to solve or require different exploration strategies. This moves beyond "black-box" optimization and offers value to algorithm designers.
- **Scalability:** The framework is lightweight. Once trained, the regression model has negligible inference cost, making it highly practical for real-world deployment compared to complex online adaptive mechanisms that might slow down the search.

## ► Relevance

## Assessment: Highly Relevant

This paper is an excellent fit for the symposium, particularly for tracks dealing with **Metaheuristics**, **Parameter Tuning**, **Evolutionary Computation**, or **Machine Learning for Combinatorial Optimization**.

- **Intersection of Fields:** The paper sits squarely at the popular intersection of Machine Learning (ML) and Operations Research (OR). It uses ML not to solve the problem directly, but to configure a traditional metaheuristic, which is a core theme in modern Computational Intelligence.
- **Focus on E-CVRP:** The Electric VRP is a trending topic due to the global push for green logistics. Improvements to algorithms for this specific problem variant are highly relevant to both academic researchers and industry practitioners attending the conference.
- **Automated Design:** The work aligns with the growing trend of "Automated Algorithm Design" and "AutoML" applied to optimization, which is a staple of this symposium's scope.

## ► Rigour and Soundness

### Overview

The paper is scientifically rigorous. The experimental design is careful, acknowledging the risk of overfitting and the stochastic nature of the algorithms.

### Strengths (Soundness of Approach):

1. **Correct Choice of ML Model:** Given the small dataset size (41 instances), the decision to use **Ridge Regression** (linear model with regularization) instead of complex Deep Neural Networks is scientifically sound. It prevents overfitting and allows for the interpretability analysis presented in Section V-D.
2. **Proper Validation Protocol:**
  - The authors strictly separate the dataset into **Training** (33 instances) and **Held-out Test** (8 instances).
  - They use **5-fold cross-validation** for model selection.
  - They employ the **Wilcoxon signed-rank test** for statistical significance, which is the standard for comparing randomized algorithms.
3. **Robust Label Generation:** The two-stage offline tuning process (Coarse  $\rightarrow$  Fine) using IRACE is a rigorous method to generate "ground truth" labels. Handling potential multi-modality in parameter space (where multiple settings might be equally good) shows attention to detail often missed in similar studies.
4. **Baseline Fairness:** The comparison is made against a **Globally Tuned** configuration (also found via IRACE). This is the strongest possible baseline. Many weaker papers compare against "default" parameters; comparing against a globally optimized set proves the specific value of *instance-awareness*.

## ► Structure and Readability

The paper is exceptionally well-written and structured. The level of polish is high, suggesting experienced authors.

- **Clarity:** The problem is defined clearly. The distinction between the *offline training phase* and the *online inference phase* is articulated well (aided effectively by Figure 2).
- **Flow:** The progression from Introduction  $\rightarrow$  Background  $\rightarrow$  Framework  $\rightarrow$  Feature Engineering  $\rightarrow$  Experiments is logical and standard.
- **Visuals:**
  - **Figure 2** provides a clear architectural overview.
  - **Tables:** The tables are well-formatted. Table II (Features) is comprehensive and crucial for reproducibility.
- **Language:** The academic English is professional, concise, and free of distracting errors. The abstract is punchy and clearly summarizes the contribution and results.

### Invite to TEVC

no

If so, why and what extensions would you recommend?

Review to TEVC

► Other Comments for Authors

None

Scored Review Questions

|              | rec ⓘ      | conf ⓘ                 | oral ⓘ     | bp ⓘ   | indust ⓘ |
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Committee Comments

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Committee Comments for Authors

None